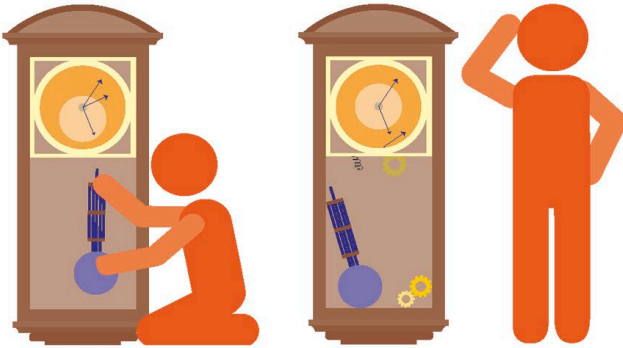


Regression tests

Sometimes, in the process of fixing a machine, another part may get broken. Regression tests are run after each new upgrade to make sure that the original software is in proper condition. They involve re-running all the unit tests and integration tests one after another. Since regression tests can take a long time to run, test cases must be prioritized. Updated files are tested first, followed by critical systems that are more likely to crash.



△ Causing new problems

Fixing one problem sometimes creates other problems elsewhere. Regression tests are designed to make sure that all of the code works as intended.

User acceptance testing

No simulation can compare to humans testing the code. The last phase of the testing life cycle is user acceptance testing, where groups of people are handed an almost finished product and asked for feedback. These comments are handed back to the developers so they can fix any issues the human testers find.

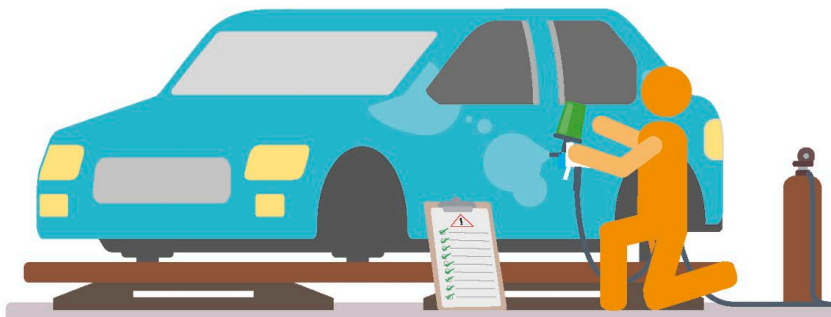


△ Alpha and beta testing

User acceptance testing has multiple rounds. Alpha testing is done internally, typically by other developers. Beta testing, or field testing, is done with a select group of external customers or volunteers.

What to do with an error?

After a member of the testing team finds an error, they file a bug report. The report includes a description of the problem, the error message, and instructions on how to reproduce the bug. Sometimes an error is non-reproducible, which means the conditions needed to trigger the bug are too hard or too random to recreate. In this case, the bug is put on hold until more information comes up. Developers are periodically assigned to comb through the list of bugs and fix them.



IN DEPTH

Testing GUIs

Testing graphical user interfaces (GUIs) is tricky. One solution is to use automation software. These programs record a developer interacting with a webpage, and then re-run the recording on command. If an update caused a button to be moved halfway across the page, the recording won't be able to finish, and the error is detected.

◁ Issue tracking

Most teams use issue tracking software to keep track of bug reports. This allows them to prioritize bugs by severity, and tick them off as they are fixed.